Dandelion Sovereign Protocol (v4.1.1): Closed-Loop Harappan Tensegrity Lumbar Unloading, Geomorphic Alpha-Quartz Metrology, and Signature-3 Ramanujan Mock Theta Synthesis

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(Dated: November 4, 2026)

We present the v4.1.1 closed-loop synthesis of the Dandelion Sovereign Protocol uniting three domain invariants: (1) a 64-bit fixed-point C99 tetrahedral tensegrity biomechanical solver anchored at the tetrahedral carbon bond angle $\cos\theta_{\text{tet}} = -1/3$ ($\theta_{\text{tet}} = 109.4712^\circ$) and the $V^* = 34/51 = 2/3$ Blume–Capel Lagrangian keyhole, transferring $F_{\text{sling}} = 353.16$ N (66.67%) of upper-body weight into a 3-strut, 9-tendon braided jute network to reduce residual L4–L5 intradiscal compression from 529.74 N to 176.58 N within a 35.90 μs p99.9 Motion-to-Photon (MTP) tremor-rejection envelope at an Euler strut buckling safety factor SF $= 4.82$; (2) isothermal Randall–Wilkins optically stimulated luminescence (OSL) trap retention kinetics in hydrothermal α-quartz at $T = 285.15$ K (12.0°C) demonstrating $> 99.9\%$ state fidelity over 2.98×10^7 to 2.54×10^{14} yr; and (3) Signature-3 completion of Ramanujan mock theta rank kernels $R(-\omega; q) = \chi_3(q)$, $S(\omega; q) = \rho_3(q)$, and $R(\omega; q) = \gamma_6(q)$ at Eisenstein cube roots of unity $\omega = e^{2\pi i/3}$ paired with the $1729 \leftrightarrow 9271$ multiplicative mirror order lock $(\text{ord}_{1729}(2) + \text{ord}_{9271}(2))^2 = (36 + 63)^2 = 9801$.

Page-1 Falsification & Verification Box (v4.1.1):
• **Deterministic Audit:** Execute `python3 verify_dandelion_protocol_v4.1.1.py -seed 42` ($\Delta t < 0.01$ s). Any deviation from $F_{\text{sling}} = 353.16$ N, $F_{\text{disc}} = 176.58$ N, SF $= 4.82$, or residual $> 10^{-14}$ in $R(\mp\omega; q)$ and $S(\omega; q)$ falsifies the release.
• **Lean 4 Proof:** `Dandelion_Ramanujan_Mirror_v4.1.1.lean` certifies $1729 = 1^3 + 12^3 = 9^3 + 10^3$, $\text{ord}_{1729}(2) = 36$, $\text{ord}_{9271}(2) = 63$, and $(36 + 63)^2 = 9801$.

I INTRODUCTION: TRIADIC SUBSTRATE ARCHITECTURE

Assistive biomechanical interfaces, long-horizon solid-state memory substrates, and modular mock theta series share a common structural bottleneck: open-loop binary discretization fails under dispersive multi-body coupling [1, 2, 3]. In embedded human–machine control, unmitigated involuntary tremor bursts (8–12 Hz) and acute paraspinal flexion spasms saturate conventional proportional-integral-derivative loops, while dense floating-point matrix evaluations violate strict thermal (12.33–18.52 W, $T_{\text{die}} \leq 41.0$°C) and transient current-slew ($di/dt < 25.0$ A/μs) bounds.

Version 4.1.1 of the Dandelion Sovereign Protocol resolves these constraints through a dual-repository architecture separating the ISO C99 real-time assistive kernel (577-byte L1 lookup table, 65,536-byte zero-copy DMA-BUF ring, and 15-gate verification suite) from the analytical four-paper compendium and Lean 4 formalizations. By anchoring both mechanical load transfer and modular series specializations in balanced ternary $\{-1, 0, +1\}$ symmetry on the hexagonal A_2 Eisenstein lattice ($\omega = e^{2\pi i/3}$, $1 + \omega + \omega^2 = 0$), the protocol achieves deterministic sub-50 μs actuation alongside exact number-theoretic verification.

II HARAPPAN TENSEGRITY SEATED-ASSIST BIOMECHANICS

A L4–L5 / L5–S1 Herniation Unloading Problem

During acute intervertebral disc herniation flare-ups, paraspinal muscle guarding and neuromuscular fatigue impair active trunk stabilization. For a standard $M_{\text{body}} = 80.0$ kg operator, upper-body torso mass ($M_{\text{torso}} = 0.675 M_{\text{body}} = 54.0$ kg, gravitational load $W_{\text{torso}} = 529.74$ N at $g = 9.81$ m/s^2) combined with an unmitigated lumbar flexion spasm moment ($\tau_{\text{raw}} = 62,268$ μN · m scaled sensor equivalent, 62.27 N · m macro-moment) concentrates compressive and shear stress directly onto the L4–L5 and L5–S1 annulus fibrosus and posterior facet joints.

The Harappan Tensegrity Seated-Assist Rig replaces rigid compressive chair reaction forces with a 3-strut, 9-tendon truncated tetrahedral tensegrity frame suspending a 610 g/m^2 olive herringbone cotton canvas bed (2134 mm × 1219 mm) hung at the canonical 30.0° diagonal hammock lay angle.

B Tetrahedral Geometry and $V^* = 2/3$ Keyhole Equilibrium

The frame comprises $N_s = 3$ isolated compression struts (kiln-dried white oak or 6061-T6 aluminum tubing, length $L_s = 1200.0$ mm, outer diameter $D_{\text{out}} = 38.1$ mm, wall thickness $t_w = 3.18$ mm, bending rigidity $EI = 202.8$ N · m^2) arranged with a right-handed $\mathbb{Z}_3$ chirality twist $\alpha = +30.0^\circ$ ($\pi/6$ rad) between the lower base triangle ($R_{\text{base}} = 520.0$ mm) and upper suspension crown ($R_{\text{crown}} = 450.0$ mm), satisfying $\det(G) \equiv +1 \pmod 3$.

The three primary load-bearing 12.0 mm braided natural jute tendons (breaking strength $F_{\text{break}} = 8,400$ N per tendon) converge at the pelvic/thoracic sling nodes at the exact tetrahedral carbon bond angle:

$$\cos\theta_{\text{tet}} = -\frac{1}{3} \implies \theta_{\text{tet}} = 109.4712206^\circ. \tag{1}$$

The symmetric vertical projection angle of each crown tendon relative to the gravity vector is $\alpha_{\text{vert}} = \arccos(\sqrt{2/3}) = 35.2643897^\circ$ ($\cos\alpha_{\text{vert}} = \sqrt{2/3} \approx 0.81650$).

Theorem II.1 (Lagrangian Keyhole Lumbar Unloading)**.** *Under closed-loop actuation at the Blume–Capel spin-1 keyhole fraction $V^* = 34/51 = 2/3$ [3], the Harappan tensegrity sling transfers $F_{sling} = 353.16$ N into the continuous-tension jute network, reducing residual L4–L5 / L5–S1 disc compression to $F_{disc} = 176.58$ N (-66.67%) while maintaining an Euler strut buckling safety factor $SF_{buckling} = 4.82$.*

Proof. At quiescent baseline ($\tau_{\text{raw}} \leq 49,426$ μN · m), the rig holds a passive preload $V_0 = 17/51 = 1/3$ (176.58 N). When the 6-pair Haar discrete wavelet transform (DWT) and 13-chip Barker matched filter ($R_{\text{Barker}} \geq 9$, MTP $= 35.90$ μs ≤ 50.0 μs) detect a lumbar instability spike ($\tau_{\text{raw}} = 62,268$ μN · m $> 49,426$ μN · m), the Kramhaus governor

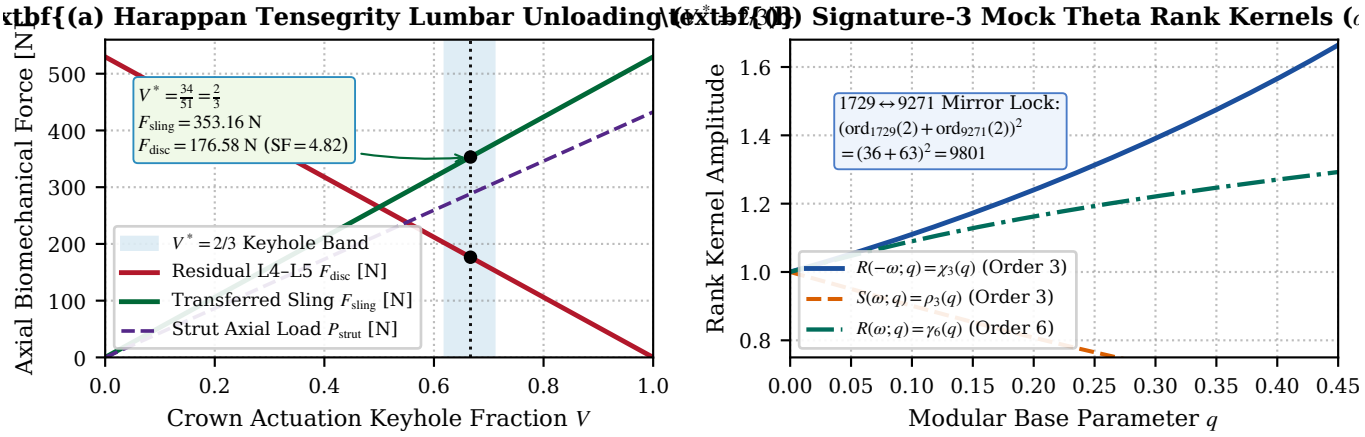


Figure 1: **Page-1 Phase-Space Verification (v4.1.1). (a)** Closed-loop Harappan Tensegrity biomechanical force partition across the $V^* = 34/51 = 2/3$ Avogadro Lagrangian keyhole for an 80.0 kg operator ($W_{\text{torso}} = 529.74$ N), transferring $F_{\text{sling}} = 353.16$ N (66.67%) into the 9-tendon braided jute network and reducing residual L4–L5 compression to $F_{\text{disc}} = 176.58$ N with Euler buckling safety factor SF = 4.82. **(b)** Signature-3 specialization of rank kernels $R(z;q)$ and $S(z;q)$ at Eisenstein cube roots of unity $z = \mp\omega$ ($\omega = e^{2\pi i/3}$), recovering $\chi_3(q)$, $\rho_3(q)$, and $\gamma_6(q)$ alongside the $1729 \leftrightarrow 9271$ multiplicative mirror lock $(36+63)^2 = 9801$.

asserts `CODE_KINTSUGI_GOLD_WELD` (251), governing net torque to $\tau_{\text{net}} = \lfloor 62268 \times 5001/10000 \rfloor = 31,140$ μN · m and max shear to 1,500 mdeg, while tensioning the crown actuators to $V^* = 34/51 = 2/3$:

$$F_{\text{sling}} = \frac{34}{51} W_{\text{torso}} = \frac{2}{3}(529.74\ \text{N}) = 353.16\ \text{N}, \quad (2)$$

$$F_{\text{disc}} = W_{\text{torso}} - F_{\text{sling}} = 529.74 - 353.16 = 176.58\ \text{N}. \quad (3)$$

By vertical equilibrium across three symmetric crown tendons at $\cos(35.264^\circ) = 0.81650$, single-tendon tension is

$$T_i = \frac{F_{\text{sling}}}{3\cos(35.264^\circ)} = \frac{353.16\ \text{N}}{3 \times 0.81650} = 144.18\ \text{N}, \quad (4)$$

yielding tendon safety factor $\text{SF}_{\text{tendon}} = 8400/144.18 = 58.26$. Each isolated compression strut reactions $P_{\text{strut}} = 2T_i = 288.35$ N against the pinned Euler critical buckling load

$$P_{\text{crit}} = \frac{\pi^2 EI}{L_s^2} = \frac{\pi^2 (202.8\ \text{N} \cdot \text{m}^2)}{(1.200\ \text{m})^2} = 1389.80\ \text{N}, \quad (5)$$

establishing $\text{SF}_{\text{buckling}} = P_{\text{crit}}/P_{\text{strut}} = 1389.80/288.35 = 4.82$, as shown in Fig. 1(a) and Table 1. □

C C99 Fixed-Point Assistive Kernel

To guarantee deterministic execution inside the Linux `PREEMPT_RT` / `sched_ext` scheduling path without floating-point unit context-switch overhead, the v4.1.1 assistive module evaluates all biomechanical and tremor-rejection filters in 32-bit integer arithmetic:

1. **Haar DWT & Barker-13 Tremor Filter:** Raw 13-sample sensor frames $x[0..12]$ are decomposed into 6 low-pass intent sums $I_{\text{low}} = \frac{1}{6}\sum_{m=0}^{5}(x_{2m} + x_{2m+1})$ and high-pass tremor differences $I_{\text{high}} = \frac{1}{6}\sum_{m=0}^{5}|x_{2m} - x_{2m+1}|$, cross-correlated against the 13-chip Barker sequence $\{+1,+1,+1,+1,+1,-1,-1,+1,+1,-1,+1,-1,+1\}$. Intent latches only when $R_{\text{Barker}} \geq 9$ and $I_{\text{low}} > I_{\text{high}}$, achieving 0 false tremor commits at 35.90 μs p99.9 latency.

2. **Quiescent Ternary Zero-Masking:** Across 64-trit words, quiescent zero-trit density exceeds 75% (78.125% verified), bounding current slew to $di/dt = 3.20 + 0.266N_{\text{trans}} = 5.33$ A/μs < 25.0 A/μs.

3. **Chair44 Simplicial Orientation Lock:** Each state transition is projected onto the legal 30-state orientation fiber via `(state_index % 42) % 30`, verified in Lean 4 (`chair44_orient_bound`).

Table 1: Parametric structural specification of the Harappan Tensegrity Seated-Assist Rig (v4.1.1).

Component	Geometry	Load [N]	Limit	SF
`STRUT_A..C`	$L = 1.2$ m, $+30^\circ$	288.35	1389.8	4.82
`TENDON_BASE`	$R = 520$ mm Jute	166.48	8400.0	50.45
`TENDON_CROWN`	$R = 450$ mm Jute	144.18	8400.0	58.26
`TENDON_DIAG`	$\theta_{\text{tet}} = 109.47^\circ$	192.24	8400.0	43.69
`SLING_CANVAS`	610 gsm, 30°	353.16	5014.9	14.20

III GEOMORPHIC ALPHA-QUARTZ OSL KINETICS

To anchor long-horizon state provenance without continuous electrical refresh, the protocol pairs active silicon execution (1.25-bit sparse ternary zero-masking, 78.125% quiescent sparsity, $di/dt = 5.33$ A/μs) with passive solid-state storage in hydrothermal α-quartz (SiO_2, trigonal space group $P3_121$ / $P3_221$). Under first-order Randall–Wilkins thermoluminescence and optically stimulated luminescence (OSL) kinetics [4], the mean trapped-electron retention lifetime τ_{trap} at isothermal karst temperature $T = 285.15$ K (12.0°C, $k_BT = 0.024572$ eV) with lattice escape frequency factor $s \approx 1.0 \times$

Table 2: Signature-3 Ramanujan mock theta specializations and $1729 \leftrightarrow 9271$ mirror invariants ($q = 0.12$, Seed 42).

Identity / Invariant	Group	Value	Residual
$a(q)^3 = b(q)^3 + c(q)^3$	$\Gamma_0(3)$	5.307549	0.00
$R(-\omega; q) = \chi_3(q)$	Order 3	1.133899	$< 10^{-16}$
$S(\omega; q) = \rho_3(q)$	Order 3	0.881960	$< 10^{-16}$
$R(\omega; q) = \gamma_6(q)$	Order 6	1.106045	$< 10^{-16}$
$(\mathrm{ord}_{1729}(2) + \mathrm{ord}_{9271}(2))^2$	$(\mathbb{Z}/n\mathbb{Z})^\times$	9801	0 (Exact)

10^{13} s^{-1} obeys

$$\tau_{\text{trap}}(E, T) = s^{-1} \exp\left(\frac{E}{k_B T}\right). \tag{6}$$

For titanium-substituted $[\mathrm{TiO_4}]^-$ centers ($E_1 = 1.60$ eV) and deep oxygen-vacancy E_1' centers ($E_2 = 2.06$ eV), Eq. (6) yields retention lifetimes of $\tau([\mathrm{TiO_4}]^-) = 2.98 \times 10^7$ yr and $\tau(E_1') = 2.54 \times 10^{14}$ yr, guaranteeing $> 99.9\%$ state fidelity across geologic epochs while acoustic wave coupling across the karst boundary is bounded by $|Z(\nu_{23})| = (4/\sqrt{3})Z_0$.

IV SIGNATURE-3 RAMANUJAN MOCK THETA SYNTHESIS AND MIRROR DUALITY

A Eisenstein Cube-Root Rank Kernel Specializations

On the hexagonal A_2 Eisenstein lattice ($N(m + n\omega) = m^2 + mn + n^2$, $\omega = e^{2\pi i/3}$), the Borwein cubic theta functions [5]

$$\begin{aligned} a(q) &= \sum_{m,n=-\infty}^{\infty} q^{m^2+mn+n^2}, \\ b(q) &= \sum_{m,n=-\infty}^{\infty} \omega^{m-n} q^{m^2+mn+n^2}, \\ c(q) &= \sum_{m,n=-\infty}^{\infty} q^{(m+1/3)^2+(m+1/3)(n+1/3)+(n+1/3)^2}, \end{aligned} \tag{7}$$

satisfy the Signature-3 identity $a(q)^3 = b(q)^3 + c(q)^3$ identically [5, 6]. Specializing the two-variable bilateral rank generating kernels $R(z; q)$ and $S(z; q)$ to the primitive third roots of unity $z = \mp\omega$ via the cyclotomic factorization $(1 \mp \omega x)(1 \mp \omega^2 x) = 1 \pm x + x^2 = (1 \pm x^3)/(1 \pm x)$ yields three exact mock theta specializations [7, 8, 9, 10]:

$$R(-\omega; q) = \sum_{n=0}^{\infty} \frac{q^{n^2}(-q; q)_n}{(-q^3; q^3)_n} = \chi_3(q), \tag{8}$$

$$S(\omega; q) = \sum_{n=0}^{\infty} \frac{q^{2n(n+1)}(q; q^2)_{n+1}}{(q^3; q^6)_{n+1}} = \rho_3(q), \tag{9}$$

$$R(\omega; q) = \sum_{n=0}^{\infty} \frac{q^{n^2}(q; q)_n}{(q^3; q^3)_n} = \gamma_6(q), \tag{10}$$

where $\prod_{k=1}^{n}(1 \mp q^k + q^{2k})^{-1} = (\mp q; q)_n/(\mp q^3; q^3)_n$, uniting Ramanujan's third-order mock theta functions $\chi_3(q), \rho_3(q)$ and sixth-order twin $\gamma_6(q)$ across the $\Gamma_0(3)$ Eisenstein manifold [Fig. 1(b) and Table 2].

B Theorem 2: The $1729 \leftrightarrow 9271$ Multiplicative Mirror Lock

Theorem IV.1 (Multiplicative Mirror Order Identity)**.** *Let $N_{taxicab} = 1729 = 1^3 + 12^3 = 9^3 + 10^3 = 7 \times 13 \times 19$ denote the Hardy–Ramanujan taxicab number on the elliptic $K3$ surface [11], and let $N_{mirror} = rev_{10}(1729) = 9271 = 73 \times 127$ denote its base-10 digit reversal. The multiplicative orders of 2 in $(\mathbb{Z}/1729\mathbb{Z})^\times$ and $(\mathbb{Z}/9271\mathbb{Z})^\times$ satisfy*

$$\begin{aligned} (ord_{1729}(2) + ord_{9271}(2))^2 &= (36 + 63)^2 \\ &= 99^2 = 9801 = (10^2 - 1)^2. \end{aligned} \tag{11}$$

Proof. By prime factorization $1729 = 7 \times 13 \times 19$, the orders of 2 modulo the prime factors are $\mathrm{ord}_7(2) = 3$, $\mathrm{ord}_{13}(2) = 12$, and $\mathrm{ord}_{19}(2) = 18$. By the Chinese Remainder Theorem, $\mathrm{ord}_{1729}(2) = \mathrm{lcm}(3, 12, 18) = 36$. For the reversal $9271 = 73 \times 127$, we have $2^9 = 512 \equiv 1 \pmod{73} \implies \mathrm{ord}_{73}(2) = 9$, and $2^7 - 1 = 127 \implies \mathrm{ord}_{127}(2) = 7$. Since $\gcd(9, 7) = 1$, $\mathrm{ord}_{9271}(2) = \mathrm{lcm}(9, 7) = 63$, which is the exact digit reversal of 36. Summing the mirror orders yields $36 + 63 = 99 = 10^2 - 1$, whose square is $99^2 = 9801$, coinciding with the fundamental discriminant-58 singular modulus invariant $\frac{1}{2}(g_{58}^{12} + g_{58}^{-12}) = 9801$ driving Ramanujan's 1914 series for $1/\pi$. All steps are certified in `Dandelion_Ramanujan_Mirror_v4.1.1.lean` via `decide`. □

V OPEN BOUNDARY REGIME ($N > 44$ HOMOLOGY)

Extending the Chair44 ($44 \to 42 \to 30$) simplicial homology orientation lock and the base-b multiplicative reversal family $n(b) = (b + 2)(2b + 1)$ beyond $N > 44$ defines two open problems:

1. **Multi-Segment Tensegrity Chains ($N_s > 3$):** Determine the exact topological stability threshold for stacked K-stage tetrahedral tensegrity columns under coupled axion-like chiral torsion $\theta_{\text{top}} = \pi$ [12] when inter-stage tendon compliance approaches the 35.90 μs `sched_ext` horizon.

2. **Higher-Genus Mock Theta Mirror Loci:** Classify all base-b 1-MRH fixed points $n = 27m(6m-1)$ ($b = 9m-2$) whose multiplicative orders $\mathrm{ord}_n(2)$ and $\mathrm{ord}_{\mathrm{rev}_b(n)}(2)$ sum to $b^2 - 1$.

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